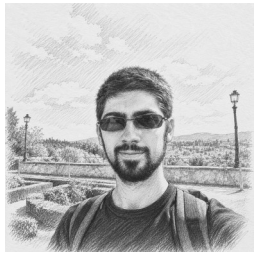
Pseudorandom Functions in $AC^0[2]$ in the Bounded-Query Setting



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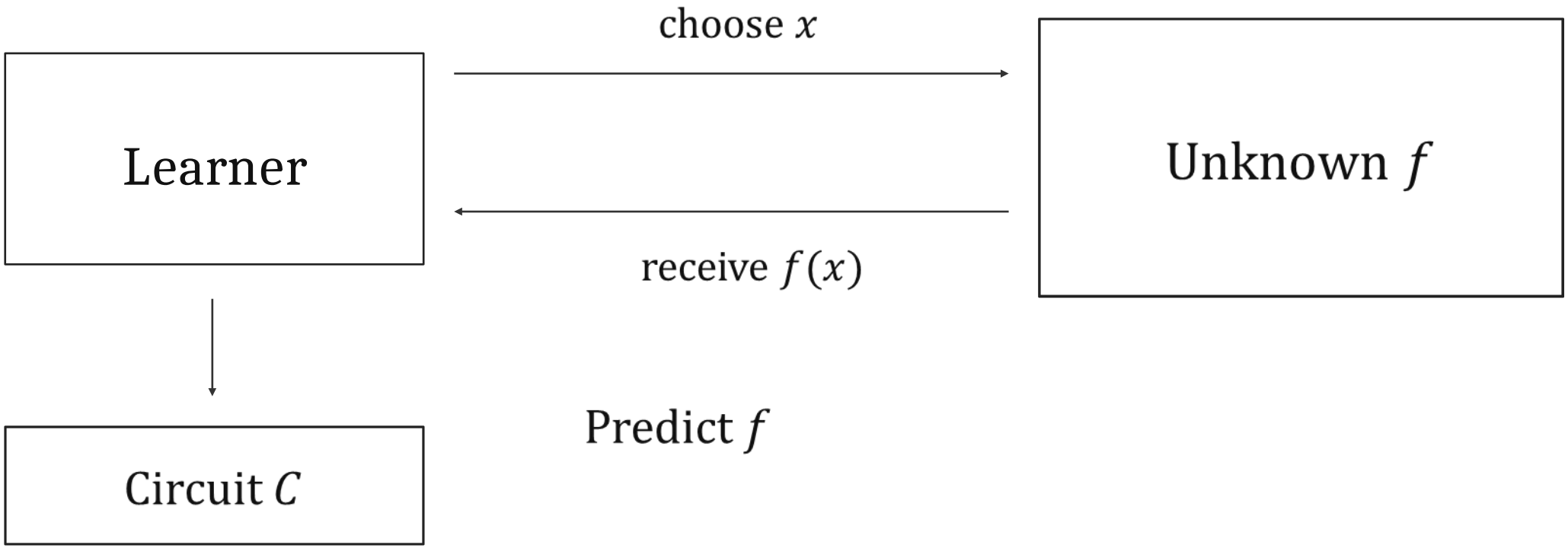


Peter Scholl

Aarhus

Learning in $\mathrm{AC}^0[p]$

What is learning?



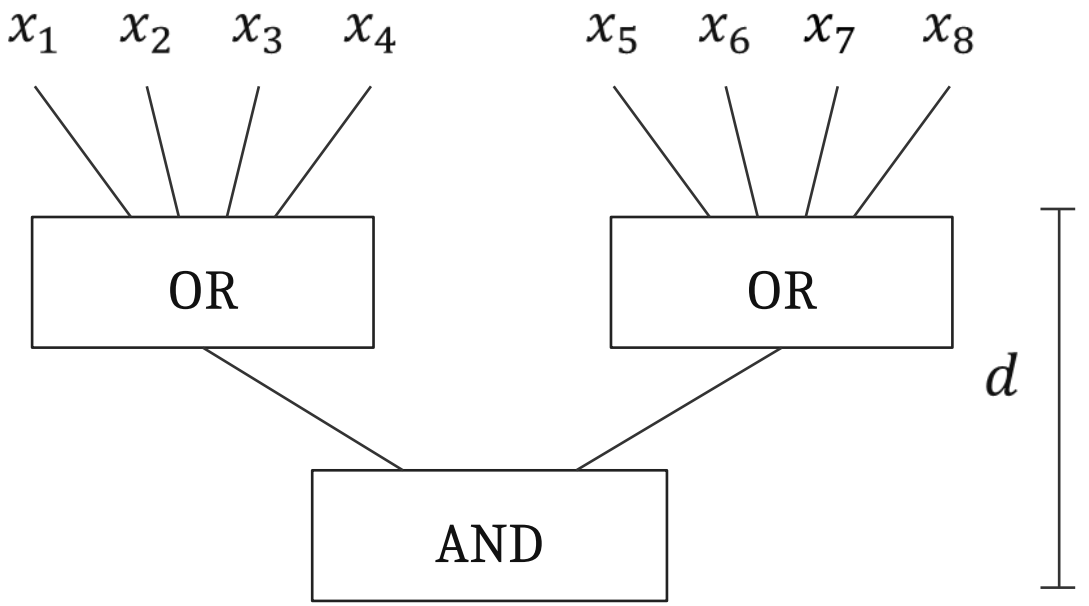
What is AC^0?

Polynomially many gates

AND, OR, NOT

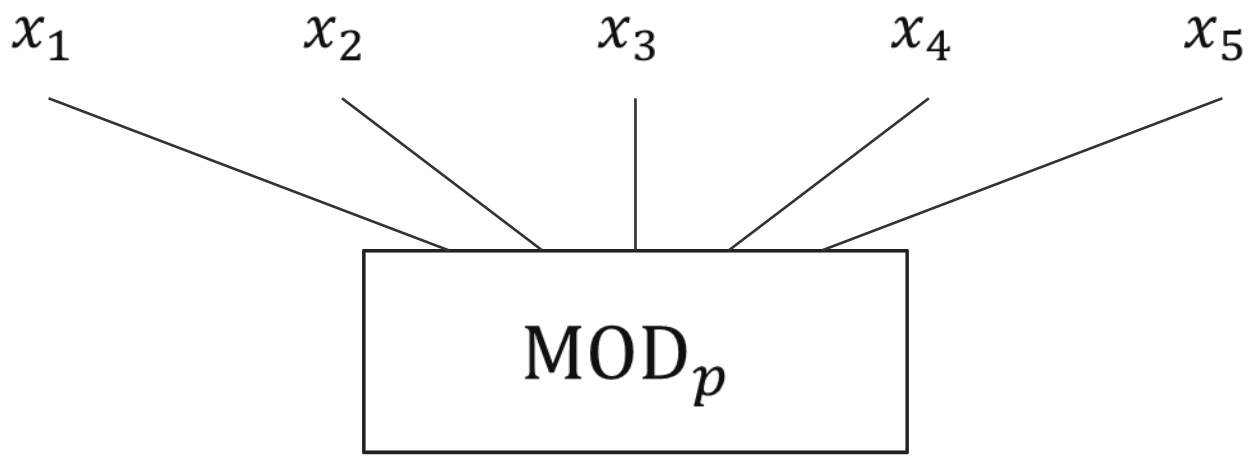
Unbounded fan-in

Constant depth



input length n; depth d.

What is $\mathrm{AC}^0[p]$?

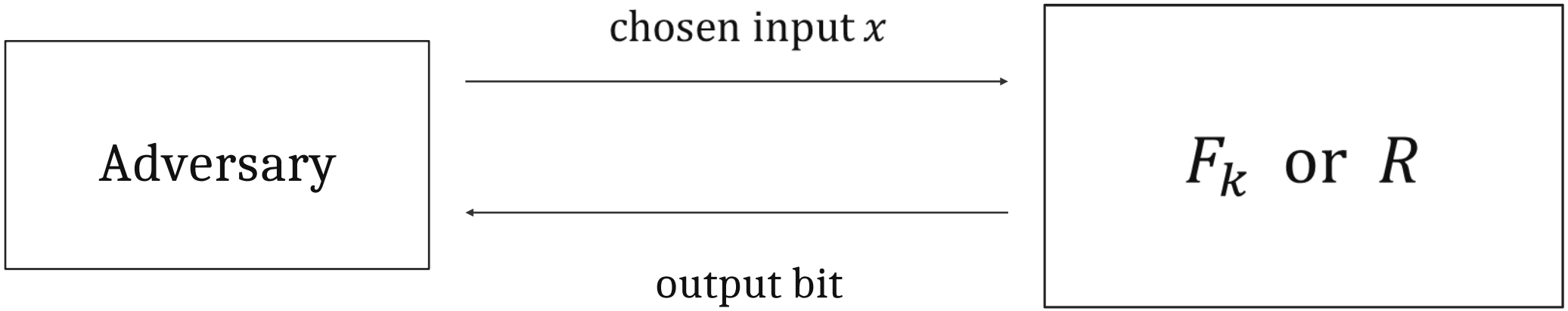


$$\mathrm{MOD}_p(x) = 1 \quad \text{iff} \quad \sum_i x_i \equiv 0 \pmod{p}$$

Why care about $\mathrm{AC}^0[p]$?

Class	Learning / distinguishing	Pseudorandom Function (PRF) landscape
AC^0	Quasipoly-time learning [LMN89]; 2-query distinguisher	
$\mathrm{AC}^0[p]$ **(p prime)**	Quasipoly-time learning [CIKK16] and distinguisher [Razborov87, Smolensky87]	
$\mathrm{AC}^0[pq]$ **($p \neq q$ prime)**		
TC^0: Contains threshold / majority gates		

Pseudorandom functions (PRFs)



F_k: efficient; secret uniform key k

R: a uniformly random function

Why care about $AC^0[p]$?

Class	Learning / distinguishing	Pseudorandom Function (PRF) landscape
AC^0	Quasipoly-time learning [LMN89]; 2-query distinguishing test	
$AC^0[p]$ **(p prime)**	Quasipoly-time learning [CIKK16] and distinguishing [Razborov87, Smolensky87]	PRFs from Factoring, DDH, and Ring-LWE with *quasipolynomial security*
$AC^0[pq]$ **($p \neq q$ prime)**		Dark Matter [BPISW18]: PRF candidate in $AC^0[6]$ with conjectured *exponential security*
TC^0: Contains threshold / majority gates		PRFs from Factoring, DDH [Naor-Reingold04], and Ring-LWE [Banerjee-Peikert-Rosen08] with *subexponential security*

How far can we push learning for $\mathrm{AC}^0[p]$?

Can we obtain a lower-query learning algorithm?

This work: Conjecture: No.

This work: Conjecture

For each fixed prime p, there exists $c > 0$ such that

$\mathbf{AC^0[p]}$ **cannot be learned** with:

Membership queries

$$Q = \kappa^{(\log \kappa)^c}$$

Running time

$$T = 2^{\kappa^c}$$

Are PRFs in $\mathrm{AC}^0[pq]$ the best we can hope for?

Can we remove the need to mix different moduli?

Keep the three resources separate

$$|\Pr[A^{F_k} = 1] - \Pr[A^R = 1]| \leq \varepsilon$$

Q	T	ε
oracle queries	computation time	distinguishing advantage

What if queries are bounded?

$$Q \ll T$$

Can $\mathrm{AC}^0[p]$ support subexponential time security in this regime?

Applications

Secure messaging

Double Ratchet: unique message keys and deletion after use

FHE transciphering

Encrypt each message under a new key

TLS

Per-key usage limits

Private Set Intersection (PSI) from Oblivious PRFs (OPRFs)

Use a new OPRF key for each PSI run

What if queries are bounded?

$$Q \ll T$$

Can $\mathrm{AC}^0[p]$ support subexponential time security in this regime?

Example

PRF from LPN in almost-$\mathrm{AC}^0[2]$

Candidate is depth d with subexponential security
when limited to $n^{d/3}$ queries

This work: a PRF candidate in $\mathrm{AC}^0[2]$

$$F_k \in \mathrm{AC}^0[2]$$

$$Q = \kappa^{\log^c \kappa}$$

quasipolynomial

$$T = 2^{\kappa^c}$$

subexponential

$$\varepsilon \leq 1/T$$

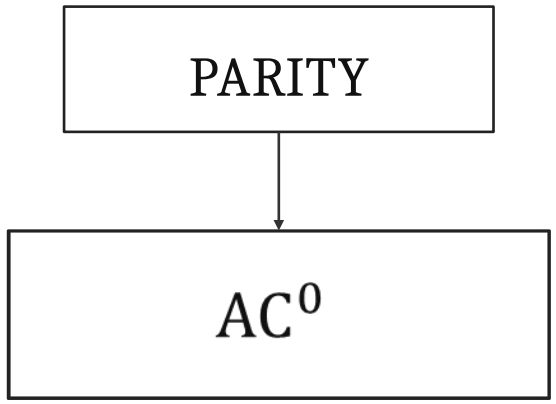
subexponential

For some fixed $c > 0$, with κ the key length.

This work

- **Conjecture:** Hardness of Low-Query Learning for $\mathrm{AC}^0[p]$
- **Conjecture & Cryptanalysis:** A PRF candidate in $\mathrm{AC}^0[2]$
- **Theorem:** No PRFs in $\mathrm{MOD}_2 \circ \mathrm{AC}^0$
- **Theorem:** A universal PRF in $\mathrm{AC}^0[2]$

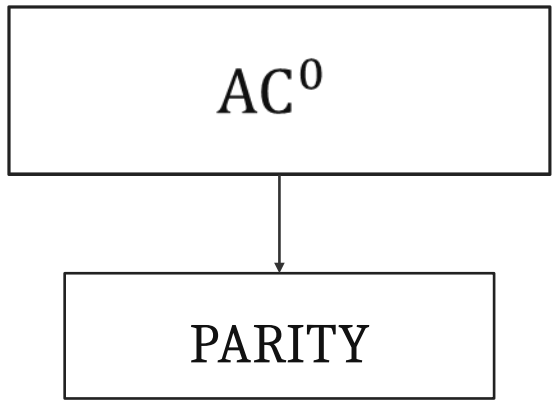
Where should the parity gates go?



[ABGKR14] conjecture

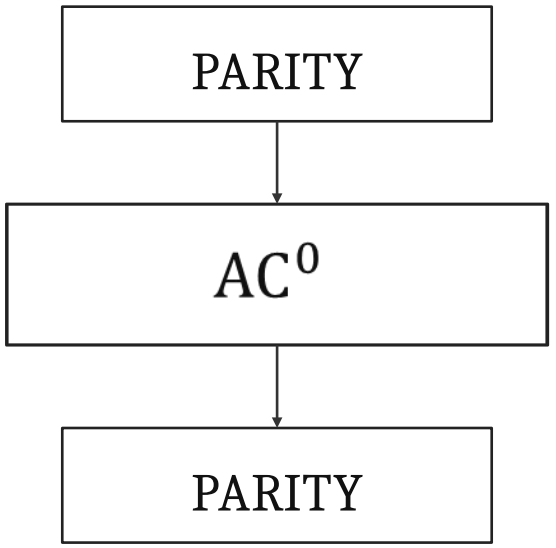
Heavy Fourier coefficient
$\Longrightarrow$ chosen-query attack

Open



Our theorem

No PRFs:
polynomial-query
distinguisher



Our candidate

$\mathrm{MOD}_2 \circ AC^0 \circ \mathrm{MOD}_2$

Starting point [Akavia-Bogdanov-Guo-Kamath-Rosen14]

Broken candidate **weak** PRF in $AC^0[2]$

Weak: Adversary only gets access to random examples

$$\mathrm{TRIBES}(y) = \bigvee_{j=1}^{m} \bigwedge_{k=1}^{n} y_{j,k}$$

TRIBES chosen to be inapproximable by low-degree $\mathbb{F}_2$ polynomials.

$$f_{s,A}(x) = \langle x, s \rangle \oplus \mathrm{TRIBES}(Ax)$$

Balanced TRIBES: $m \asymp M/\log M,\ n \asymp \log M,\ M = mn.$

Starting point [Akavia-Bogdanov-Guo-Kamath-Rosen14]

Broken in quasipolynomial time by a rational degree attack [Bogdanov-Rosen17]

Need $f(0)$ to be random

$$\mathrm{TRIBES}(y) = \bigvee_{i=1}^{m} \bigwedge_{k=1}^{n} y_{j,k}$$

$$f_{s,A}(x) = \langle x, s \rangle \oplus \mathrm{TRIBES}(Ax)$$

Balanced TRIBES: $m \asymp M/\log M,\ n \asymp \log M,\ M = mn.$

Avoiding rational degree attacks

[Boyle-Couteau-Ishai-Gilboa-Kohl-Scholl21]

Candidate **weak** PRF in $AC^0[2]$

$$g(x) = \bigwedge_{k=1}^{\lambda} \bigvee_{i=1}^{\lambda} \bigwedge_{j=1}^{\log \lambda} x_{kij}$$

$$f_{s,A}(x) = \langle x, s \rangle \oplus g(Ax)$$

Good weak PRF: Increases rational degree to $O(\lambda)$

Insufficient as a strong PRF: Has heavy Fourier coefficients → learnable by Kushilevitz-Mansour with chosen queries

Our candidate

Candidate strong PRF in $AC^0[2]$

$$g(x) = \bigoplus_{k=1}^{\lambda} \bigvee_{i=1}^{\lambda} \bigwedge_{j=1}^{\log \lambda} x_{kij}$$

$$f_{A,b}(x) = g(Ax + b)$$

A is a randomly chosen invertible matrix
b is random

Cryptanalysis

Attack	Established property
Rational-degree attacks	$\mathrm{rdeg}(F) \geq \Theta(\lambda)$
Heavy Fourier coefficients	$\max_s \lvert\hat{f}(s)\rvert \leq 2^{-\Omega(\lambda \log \lambda)}$
Linear tests	All fixed polynomial-query nonadaptive tests [Melissaris26]; additional structured adaptive cases
Sparse-polynomial learning	Subexponentially many monomials with overwhelming probability

Open: General polynomial-query adaptive linear tests

Open questions

Thank you!

1. Prove Akavia's conjecture: no PRFs exist in $\mathrm{AC}^0 \circ \mathrm{MOD}_2$
2. Construct or rule out a depth-3 PRF in $\mathrm{AC}^0[p]$
3. Can the bounded-query setting yield more efficient PRFs tailored for specific applications (i.e. Double Ratchet, TLS, FHE transciphering, PSI from OPRFs)?